INSA SUMMER CAMP 2018

Department of Development



INTERNMATCH

A two-sided internship matching platform

Fair Chances. Real Fit.

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Organization	INSA Summer Camp 2018
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GitHub	github.com/edenalemayheu
Repository	github.com/edenalemayheu/InternMatch
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InternMatch – Project Documentation

1. Overview

InternMatch is a two-sided internship matching platform where students rank the postings they want, companies rank the students they want, and a transparent algorithm pairs them by mutual preference. Its tagline is ***Fair Chances. Real Fit.***

Item	Detail
Project type	Individual graduation project, designed and built solo by Eden Alemayehu
Users	Students, companies, and one admin who controls the round
Core idea	Ranking replaces applying: a student's ordered list of postings is their single application
Matching method	Score-based greedy matching, inspired by stable-matching systems such as Gale–Shapley
Repository	github.com/edenalemayheu/InternMatch
License	MIT

2. Problem Statement

Finding an internship is an important step for every university student, because it turns classroom knowledge into real experience and often leads to a first job. Yet the way internships are usually given out today is mostly one-sided. Students send applications and wait, and companies pick from whoever happens to apply. Neither side can see the full picture, so the result is often a poor fit, wasted effort, or a missed opportunity.

Problems students face

- A student usually applies to many companies at once, writing a new application each time, and still has no idea where they truly stand.
- Students rarely get a chance to say which company they want most, so a dream choice is treated the same as a backup.
- Replies can take weeks or never come, which makes it hard to plan and leaves strong students without a placement.

- Students from smaller departments or with less experience can be overlooked simply because they applied late or less often.

Problems companies face

- Popular companies receive far more applications than they can read, so good candidates can be missed in the pile.
- Companies judge each applicant alone, without knowing how interested the student really is, or whether other companies are also offering them a place.
- Less well-known companies struggle to reach qualified students, and may end up with empty spots.
- Reviewing, shortlisting and interviewing are done by hand across emails and spreadsheets, which takes time and is easy to get wrong.

Problems with the process itself

- Results often depend on who applied first or who has the best single score, not on who fits best.
- Offers arrive at different times, so students may accept one early, then regret it when a better one comes later.
- There is no clear, shared rule that both sides can see and trust, so the outcome can feel unfair.

The need. Both students and companies need a fair, organized way to say who they truly want, and a clear method that turns those wishes into matches. This is the problem InternMatch is built to solve: instead of applying and waiting, both sides rank their choices, and the system pairs the people who want each other the most.

3. Objectives

Main objective-To build a website where students and companies each rank who they want, and the system fairly pairs them for internships.

Specific objectives

- Let students create one profile and use it in every round.
- Let students rank postings instead of sending many separate applications.
- Let companies review applicants, shortlist a few, interview them and rank the ones they met.
- Match students and postings using a clear, fair and easy-to-explain method.

- Keep student contact details private until a company has shortlisted them.
- Give an admin the tools to run, test and repeat a full round

4. Requirements

What the system must do

All users

- Sign up and log in as a student or a company, using an email and password.
- See only the pages that belong to their role, so a student cannot open company pages and the other way round.

Student

- Build a profile once (name, field of study, year, skills, portfolio and availability), and reuse it in every round.
- Browse postings that match their field and skills, or switch to see all postings.
- Rank the postings they want, from first choice downwards. This ranked list is their application.
- Receive a notification when a company shortlists them and when results are ready.
- See their match result on Match Day.

Company

- Create postings with a title, required field, required skills and a number of spots.
- See the students who ranked each posting, together with their profiles.
- Shortlist or reject each applicant, so interviews are limited to the best candidates.
- Write private interview notes for each shortlisted student.
- Rank or score the students they interviewed, to give their final preference.
- See the students matched to their postings on Match Day.

Admin

- Move the round to the next step when it is time.
- Run the matching and check who was matched and who was not.
- See all profiles, postings and rankings at any time, to check and explain what is happening.

- Reset the round to a clean start, so the same demo can be repeated.

Qualities the system must have

- Privacy: a student's email is shown only to a company that shortlisted them.
- Fairness: rankings cannot be changed after a step closes, and results appear for everyone at the same time.
- Clarity: the matching method is simple enough to explain and check by hand.
- Repeatability: the admin can reset a round so the same demo can be run again.
- Ease of use: the website works on both computers and phones, with clear messages about the current step.

5. Use Cases

Create a profile (student or company)

- After signing up, the user fills in a short profile.
- Result: they reach their own dashboard.

Browse postings (student)

- The student sees postings that fit their field, or switches to see all postings.
- Result: a short list of postings worth considering.

Rank postings (student)

- The student drags postings into order, from first choice downwards, and submits.
- Result: their application for the round is saved.

Create postings (company)

- The company adds a title, the required field, the skills it wants and the number of spots.
- Result: the postings are open for students to rank.

Review and shortlist (company)

- After rankings lock, the company reads the profile of each student who ranked it and marks them shortlisted or rejected.
- Result: a shortlist is created, chosen students are notified, and the company can now see their contact email.

Interview and rank (company)

- The company interviews the shortlisted students, writes notes, and gives a final ranking or scores.
- Result: the final rankings are ready for matching.

See the result (student and company)

- On Match Day each side opens the results page.
- Result: students see their match, or a kind message if they were not matched; companies see the students placed with them.

Run the round (admin)

- The admin moves the round through each step and runs the matching.
- Result: the results are ready to be revealed to everyone.

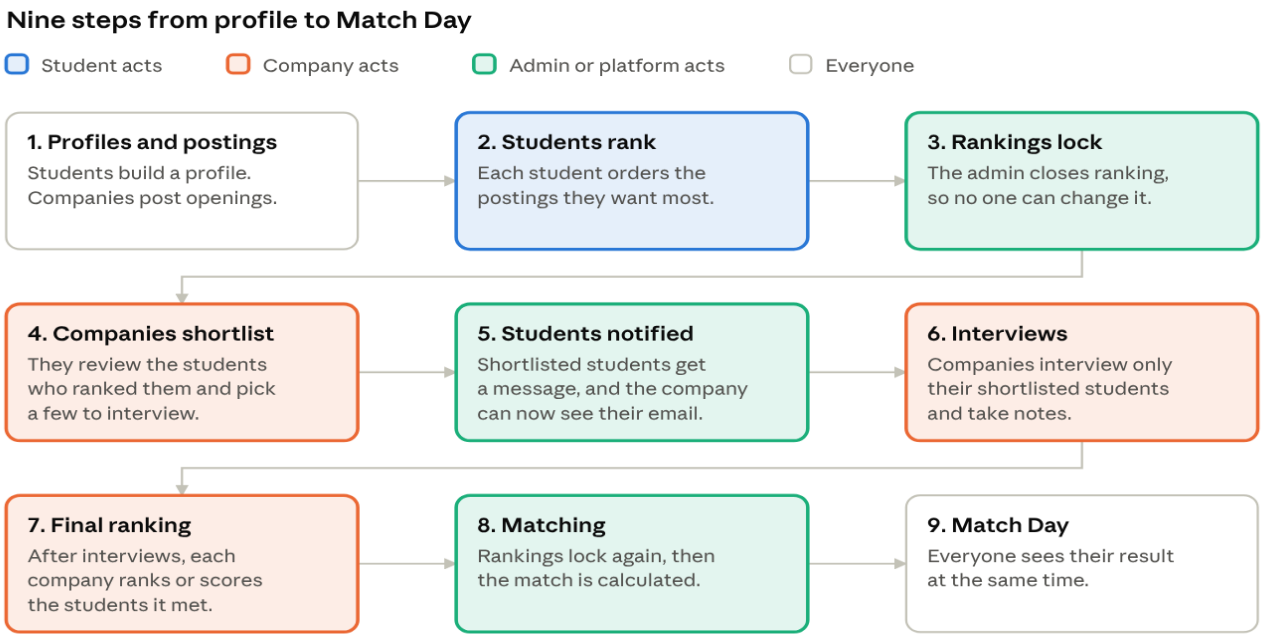
Reset the round (admin)

- The admin confirms a reset.
- Result: the round returns to a clean start, so the demo can be run again.

6. How a Round Works

Every round follows nine steps.

Shortlisting and interviews are separate on purpose: shortlisting is a quick review of everyone who ranked a company and interviews are only for the few who make the shortlist.



7. How the Match Is Made

The match uses a simple point system that anyone can check by hand. For every student and posting who picked each other, we add two numbers: 1 divided by the rank the student gave the posting, plus 1 divided by the rank the company gave the student. A first choice is worth 1 point, a second-choice half a point, a third choice a third, and so on. The higher the total, the stronger the match.

We then go down the list from the strongest pair to the weakest. If both the student and the posting are still free, they are matched. If not, we skip to the next pair. Postings can take more than one student if they have several spots.

A small example. Students A and B both rank posting X first and Y second. Company X prefers B over A, and company Y prefers A over B. Each posting has one spot.

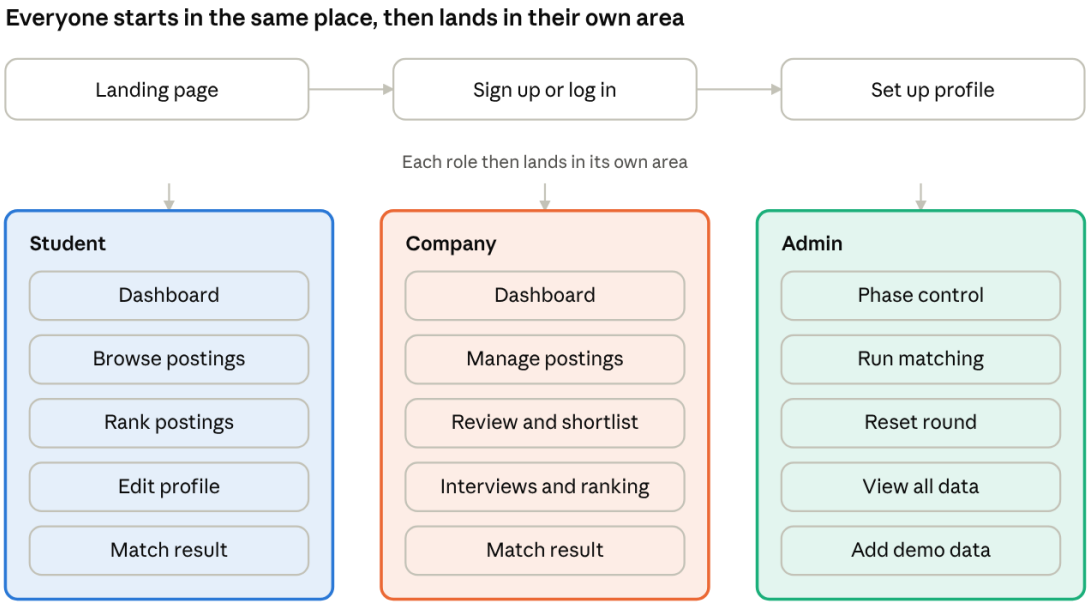
Pair	Student's rank	Company's rank	Points	What happens
B and X	1st	1st	2.0	Matched
A and X	1st	2nd	1.5	Skipped, X is already full
A and Y	2nd	1st	1.5	Matched
B and Y	2nd	2nd	1.0	Skipped, B already has a match

Result: B goes to X and A goes to Y. Both students end up where the company also wanted them.

Students who are not matched, and postings that stay empty, simply take part again in the next round.

8. What Each User Sees

Everyone enters through the same landing page and login. After that, each type of user lands in their own area of the website, which means there is role based access.

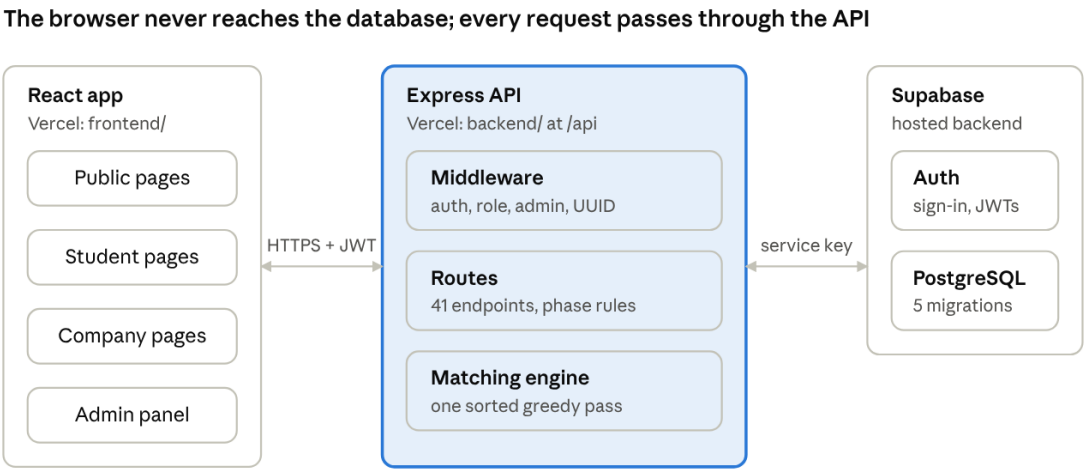


9. Tech Stacks

InternMatch is built with common and free tools. It has three layers: the website people use, a server that handles the rules, and a database that stores the information. The website never reads the database directly; it asks the server, and the server decides what each person is allowed to see.

Part	Tool used
Frontend	React with Vite
Backend	Node.js with Express
Database and sign-in	Supabase (PostgreSQL)
Hosting	Vercel
Matching	It’s own points-based method (section 4)

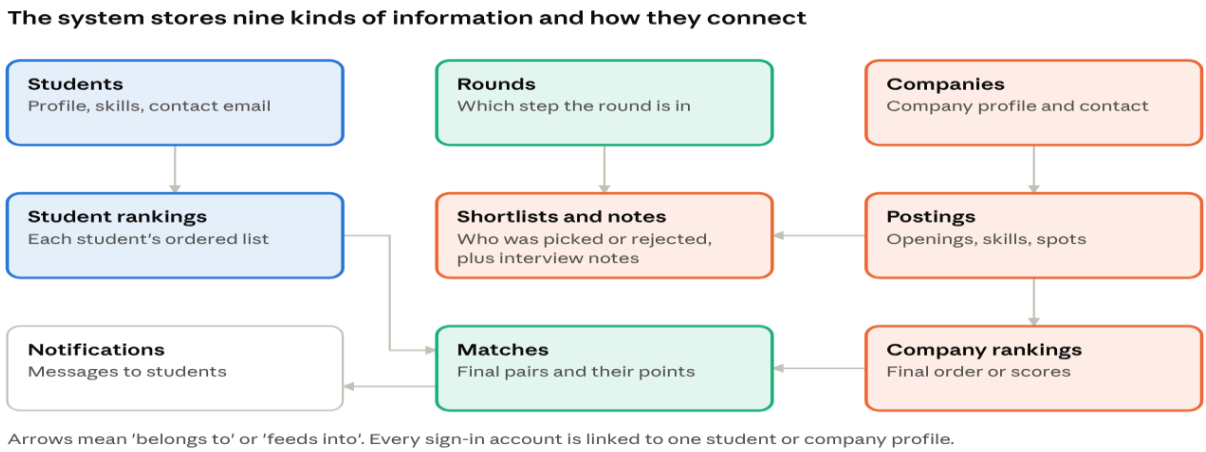
The full technical details, setup steps and database design are in the project's GitHub README.



The website never reaches the database directly. Every request goes through our server, which keeps private data safe.

10. Database Design

The database keeps everything a round needs in separate, tidy groups. The picture shows what each group holds and how the groups feed each other.



11. Testing

I tested the system at three levels: the small parts on their own, each function of the server, and a full round from start to finish.

Matching method

- How: 14 automatic tests checked the points, ties, postings with several spots, and students who stay unmatched.

- Result: all passed.

Website screens and data handling

- How: 78 automatic tests checked how the website talks to the server, including the full student and company workflow.
- Result: all passed.

Server rules

- How: automatic tests checked sign-in, profiles, permissions, actions in the wrong step, actions by the wrong role, and bad input on every page that accepts input.
- Result: all passed.

Completeness

- How: a checker confirmed that each of the 41 planned server functions exists and really works.
- Result: all 41 confirmed.

Full round

- How: a script played a whole round: sign-up, ranking, shortlisting, interviews, final rankings, matching and reveal.
- Result: completed successfully, with four matches.

Final build

- How: the website was built for publishing.
- Result: built without errors.

I also checked that the admin and student demo accounts can sign in to the real online database.

12. Limitations and Future Work

Limitations

- The admin moves the round to the next step by hand instead of by calendar date. This makes demos easy but needs a person in a real round.
- Interviews are arranged outside the website, by email or phone, once a company has a student's contact.
- Notifications appear inside the website only.

- The match is fast and easy to explain, but it is a simplified method, not a formal proof that no pair would prefer to switch.
- Students and postings that are not matched wait for the next round; there is no waiting list.

Future work

- Messaging between shortlisted students and companies inside the website.
- A short second round for students and postings left unmatched.
- Automatic step changes on set dates.
- Email alerts for shortlisting and results.
- Using the stricter Gale–Shapley method as an option, to compare results.

13. Conclusion

InternMatch shows that internship placement can be fairer and clearer when both sides say who they truly want. Students rank instead of applying many times, companies shortlist and interview only the people who stand out, and a simple points method pairs everyone on the same day. The website covers the full round for students, companies and the admin, protects private details, and can be reset and run again for demonstrations.

Building it taught me that writing the rules down first makes the work easier, that a simple method is easier to trust and explain than a complicated one, and that letting the admin control each step makes testing fast.